

Tayyip Mirza

Senior Data Engineer | Principal Data Architect | Data Science & Analytics Leader

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Summary

Senior Data Professional with 12+ years of experience across Data Engineering, Data Architecture, Analytics, and Applied Data Science. Expert in building scalable lakehouse and warehouse platforms, advanced analytics layers, and machine learning foundations that support business, product, and executive decision-making. Strong background in Python, SQL, Spark, Airflow, dbt, Snowflake, Microsoft Fabric, AWS, and modern BI ecosystems. Proven ability to translate raw data into governed analytical models, KPI frameworks, experimentation datasets, and predictive features used by analysts, data scientists, and leadership teams. Experienced in statistical modeling, machine learning pipelines, and model deployment for real-world business solutions.

Skills

Data Architecture & Modeling

Structures analytical and operational data through scalable Lakehouse patterns, dimensional models, governed schemas, and lineage-driven architecture for analytics, BI, and data science consumption.

Cloud Analytics Platforms

Delivers unified analytical environments across Microsoft Fabric, Snowflake, and AWS with scalable compute, optimized storage, secure access layers, and analytics-ready semantic models.

Storage & Data Systems

Implements structured and semi-structured storage systems using Parquet, Delta, and warehouse formats with partitioning strategies and performance-tuned access layers for reporting and modeling.

Infrastructure & Deployment Engineering

Automates infrastructure using Terraform and CI/CD pipelines, standardizes execution environments, and manages cloud resource provisioning for data, analytics, and machine learning workloads.

Data Pipeline & ML Engineering

Designs ingestion and transformation pipelines across batch and streaming workloads with reliable orchestration, CDC patterns, and analytics-friendly transformation layers. Implements feature engineering, predictive datasets, and machine learning pipelines for forecasting, classification, regression, and recommendation systems.

Orchestration & Workflow Management

Coordinates data workflows, model transformations, experiments, and deployment cycles through Airflow, dbt, Fabric pipelines, and ML workflow tools with version control and observability.

Programming & Data Processing

Builds data logic, analytical queries, feature preparation, and distributed processing components using Python, SQL, and Spark across ingestion, transformation, analytics, modeling, and ML layers. Skilled in libraries such as scikit-learn, TensorFlow, PyTorch, pandas, and NumPy.

Analytics, BI & Data Science Foundations

Develops KPI frameworks, dashboard datasets, statistical analysis, experimentation datasets, and predictive analytics foundations supporting analysts, data scientists, and leadership teams.

Professional Experience

Lead Data Engineer

01/2021 – Present

Factual

- Architect and deliver enterprise lakehouse platforms supporting analytics, BI, and machine learning workloads.
- Build batch and streaming pipelines using Spark, Kafka, Airflow, dbt, and Microsoft Fabric.
- Design governed semantic models enabling KPI dashboards, ad-hoc analytics, and ML ready datasets.
- Develop feature-ready datasets and deploy ML pipelines for forecasting, experimentation, and predictive modeling.
- Optimize Snowflake architecture including clustering, cost management, and secure access layers.
- Automate AWS infrastructure with Terraform and CI/CD pipelines.
- Partner with product, analytics, and data science teams to translate business questions into analytical and predictive datasets.

Data Engineer II

09/2017 – 12/2020

Heap Analytics

- Designed Snowflake warehouse layers for product analytics, behavioral intelligence, and ML model consumption.
- Built Spark pipelines on AWS EMR processing large-scale event streams and feature engineering.
- Developed real-time ingestion services using Kafka and AWS Lambda.
- Created analytics datasets for KPI reporting, funnels, cohorts, experimentation, and predictive modeling.
- Implemented anomaly detection, validation checks, and observability monitoring for ML pipelines.
- Supported analysts and data scientists with clean, query-optimized datasets.

Data Pipeline Engineer

05/2014 – 08/2017

Mode Analytics

- Built ELT frameworks using Snowflake, dbt, and Python for analytics and data science consumption.
- Created PySpark jobs on Databricks for analytical modeling, feature engineering, and predictive analysis.
- Enabled near real-time analytics through Kafka streaming pipelines.
- Deployed Airflow workflows with SLA monitoring, lineage tracking, and ML pipeline orchestration.
- Prepared experimentation datasets supporting A/B testing, statistical analysis, and predictive models.

Associate Data Engineer

01/2012 – 04/2014

Chartio

- Built Python and SQL ETL pipelines supporting BI, analytics, and machine learning workflows.
- Automated extraction via Airflow DAGs improving refresh reliability.
- Designed dimensional schemas for scalable business reporting and ML-ready datasets.
- Tuned Redshift performance and optimized analytical queries.
- Integrated external APIs into analytics-ready datasets for modeling and predictions.

Certificates

AWS Certified Data Analytics - Speciality

Databricks Certified Data Engineer Professional

Google Professional Data Engineer

Education

Bachelor of Science: Computer Science

School of Engineering (Grove)